A new two-beam differential resonant micro accelerometer

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***Abstract*— A novel uniaxial micro-machined resonant accelerometer is presented. The device working principle is based on the stiffness variations of a beam which is fully clamped to the substrate on one side and clamped to a seismic mass on the other side. A movement of the seismic mass, induced by an external acceleration, causes either a compressive or a tensile stress on the beam, inducing a variation of its stiffness. This variation results in a change of the resonance frequency of the beam. The accelerometer is arranged in a differential structure, with two beams built in such a way that their changes in the resonance frequency have opposite sign. This solution allows obtaining a doubled sensitivity with the same area and allows reducing the non linear behavior. First experimental results show that the device has an overall differential sensitivity Δf_{res}/g $\approx$ 450 Hz/g in the linear range of operation, with an overall area occupation lower than (500 μm)^2.**

I. Introduction

Micro Electro Mechanical Systems (MEMS) represent the emerging technology for the production of low-cost and low-power inertial sensors. In the last ten years these devices have completed the transition phase from the research to the industrial market [1]–[3]. Most of the available MEMS accelerometers is based on the movement of a seismic mass which forms a set of capacitors with suitably designed constrained parts. An external acceleration directly results in a capacitance variation, which is the electrical signal to be read. These kinds of capacitive accelerometers can be split in two classes depending on how the capacitances are formed, i.e. by means of parallel plates or by means of comb fingers. The quest for low-cost products makes it compulsory to minimize the Silicon area occupied by a device. Both parallel plates and comb fingers capacitive accelerometers suffer from this dimension scaling: parallel plates capacitors are prone to pull-in instability, which gives a limit to the minimum gap dimension between plates, and thus to the miniaturization of the accelerometer [4], [5]; comb-finger capacitors on the contrary do not suffer pull-in problems but need a doubled area to ensure the same sensitivity of parallel plates.

In order to overcome these problems and to design even smaller devices, resonating accelerometers have been proposed in the scientific literature [6]–[9], [14], [15]. The working principle is different in that an external acceleration causes a variation in the resonance frequency of a suitable micromechanical structure. The change in the resonance frequency is the signal to be read. Resonant sensing, with respect to other sensing principles, has the advantage of an immunity to pull-in instability, a high potential sensitivity and a large dynamic range. To obtain high sensitivity when using very small inertial masses, a particular care should be given to the geometry of the device.

This paper focuses on the analysis and the experimental results of the new uniaxial accelerometer proposed in [10], fabricated using the ThELMA (Thick Epitaxial Layer for Microactuators and Accelerometers) surface micro-machining process of STMicroelectronics [11]. The basic principles on which this sensor works are similar to those of already existing resonant accelerometers, but the geometrical setting and hence the properties of the mechanical part are completely different. The mechanical behavior of the resonating parts and of the whole system are described and a geometrical optimization is performed in order to obtain the highest sensitivity at a certain proof mass.

The sensitivity of the resonant accelerometers is defined as the frequency shift produced by an external acceleration of 1 g. The resonant accelerometers obtained through surface micromachining reported in the literature typically have sensitivity ranging from 40 Hz/g up to 160 Hz/g, for the quite big device proposed in [9]. The optimized new design allows to produce a very small accelerometer: the proof mass has a square shape of 400 μm x 400 μm. The experimental results show a high sensitivity of more than 430 Hz/g in a differential configuration, for devices with a quality factor Q around 200.

II. Accelerometer design

The operating scheme of a linear accelerometer can be described as follows. An inertial mass m is attached to a frame by means of a spring of stiffness k and is subject

to damping (mainly in the free molecular flow regime at typical MEMS packaging pressures) represented by a damper of coefficient b. When the reference frame is subjected to an external acceleration a, the oscillation of the inertial mass is governed by the dynamic equilibrium equation:

$$m\ddot{x} + b\dot{x} + kx = ma \tag{1}$$

If the frequency Ω of the external acceleration *a(t)* is well below the resonance, i.e. if $\Omega << f_{res}$ ($f_{res} = (1/2\pi)\sqrt{k/m}$ being the resonance frequency of the accelerometer) the accelerometer response is quasi-static and $x(t)$ = *(m/k)* · *a(t)*. The external acceleration turns out to be proportional to the mass displacement and the sensing can be done by measuring the mass displacement e.g. via the capacitance variation, as in capacitive accelerometers.

In resonant accelerometers on the contrary, the input acceleration is detected as a shift in the resonant frequency of a sensing device coupled to the proof mass. The resonant part can be the proof mass itself or, more frequently, a beam connected to it. The scheme corresponding to this second case is represented in Fig. 1, where the resonant beam is shown horizontally. The operating principle is based on the dependence of the resonant characteristic on the axial force which is present in the resonator. The external acceleration a produces a force $F = ma$ on the inertial mass m. This force produces, in turn, an axial force N in a resonating beam. For a single span beam, the frequency increases in the case of a tensile load and decreases in the case of a compressive load. Denoting by f_0 the fundamental frequency of the beam, when resonating without any axial load, the resonant frequency f of the axially loaded beam can be expressed as (cfr e.g. [12], [13]):

$$f = f_0 \cdot \sqrt{1 + \alpha \frac{N \cdot L^2}{E \cdot I}} \tag{2}$$

with

$$f_0 = \frac{c^2}{2 \cdot \pi \cdot L^2} \cdot \sqrt{\frac{E \cdot I}{\rho \cdot A}}$$

where L, A and I are the length, the cross area and the inertial momentum of the beam resonator, E is the elastic modulus, ρ is the material density, while c and α are coefficients which depend on the boundary conditions of the resonator (in the approximation of clamped-clamped beam their values are 4.730 and 0.0246 respectively).
Several accelerometers based on this operating principle have been proposed in the literature through *bulk micromachining* [14], [15] and *surface micromachining* technologies [6]–[9]. These accelerometers have a different geometry (in particular different position of the resonating beam with respect to the proof mass) which greatly affects the amplification of the axial force and hence the sensitivity of the device.

In the device proposed in [10], the proof mass is suspended by means of two springs which restrain its movement to be a uniaxial translation. The resonating part of the device is

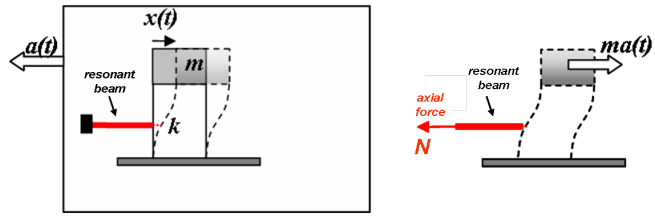
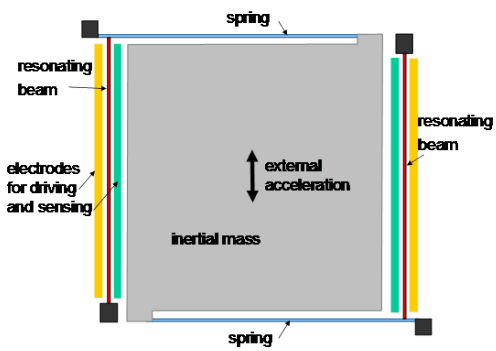


Fig. 1. Simplified scheme of a uniaxial resonant accelerometer.

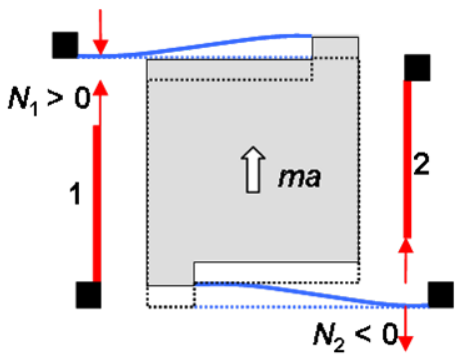
constituted by two very thin beams, attached to the substrate at one end and to the springs at the other end [see Fig. 2(a)]. The position of the resonators with respect to the anchor point of the spring is crucial to obtain high amplification of the axial force as will be discussed in the following.
Driving and sensing of the resonators is made by two parallel plate capacitances formed through electrodes attached to the substrate.
For zero external acceleration the resonators have the same nominal frequency f_0. When an external acceleration is applied, one resonator (resonator 1) is subject to a tension and the other (resonator 2) to a compression of the same magnitude, N_1 = - N_2, as shown in Fig. 2(b). The frequency f_1 of resonator 1 increases while the frequency f_2 of resonator 2 decreases. Combining the signal of the two resonators, from Eq. (2) linearized around f_0, one obtains the following frequencies difference:

$$f_1 - f_2 \approx f_0 \cdot \alpha \frac{N \cdot L^2}{E \cdot I} \tag{3}$$

Therefore the presence of two resonators undergoing opposite axial forces has several advantages: (i) the sensitivity can be doubled by measuring the difference between the frequencies of the two resonators instead of the variation of frequency of one resonator, (ii) the linearity of the system is improved, i.e. the accelerometer response can be linearised in a wider range of accelerations, (iii) the skew-symmetric geometry is less sensitive to spurious effects of thermal loading (indeed an inelastic effect causing pre-stress in the resonators is cancelled when considering the difference between the frequencies).
It is important to note that, though the resonator actuation and readout is performed through parallel plate capacitances, the resonator displacements are small compared to the gap between the resonator and the fixed plates. Moreover the effect of very large external accelerations is only a displacement of the proof mass, which results in a change of the resonator stiffness. The driving/sensing capacitances are not driven to *pull-in* even in this situation (as on the contrary may happen for parallel plate capacitive accelerometers). The relevant parameters of the device tested in this work are summarized in table 1. To allow for the detachment of the moving mass from the substrate, the inertial mass has a pattern of squared holes, so that a reduction coefficient of 0.63 should be considered when computing its effective area. Moreover due to the fabrication process, the surfaces in



(a)



(b)

Fig. 2. (a) Scheme of new resonant accelerometer; (b) effect of external acceleration.

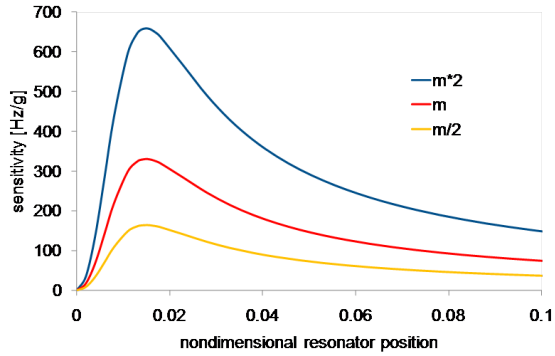


Fig. 3. Sensitivity of the accelerometer in terms of frequency variation as a function of the resonator position. Different curves correspond to the mass m of the fabricated accelerometer and of double and half the mass of this device.

TABLE I

RESONANT ACCELEROMETER PARAMETERS

Parameter	Value	Unity
Proof mass area	400 x 400	μm^2
Resonator length	400	μm
Resonator width	1.2	μm
In plane proof mass 1st mode	$\approx$ 700	Hz
Out of plane proof mass 2nd mode	$\approx$ 10000	Hz
Resonator 1st mode	$\approx$ 50000	Hz

the out of plane direction are waved, thus a resonator width sligthy reduced with respect to the nominal one should be considered.

The sensitivity of the resonant accelerometer, defined as the resonator frequency variation produced by an acceleration of 1 g, increases with the dimension of the moving mass but also strongly depends on the position of the resonating beams with respect to the anchor point of the spring. In order to reduce the device size while keeping a high sensitivity, this position was optimized by means of an analytical approach and of FEM simulations. Fig. 3 shows the sensitivity of the accelerometer, given through equation (3), as a function of the position of the resonator along the spring, normalized with respect to the spring length. One can observe that the optimal position for the resonator is very close to the anchor point of the spring at about one sixtieth of its length. The different curves correspond to different sizes of the inertial mass: the intermediate one corresponds to the tested accelerometer, with a square proof mass with a side of 400 μm. In the following sections first experimental results both on the mechanical characterization of the resonators and on the sensivitity of the accelerometer will be given.

III. MECHANICAL CHARACTERIZATION OF THE RESONATORS

The mechanical characterization of the resonant accelerometers has been performed through the low-noise, low-perturbing readout setup for MEMS capacitances described in [16]. The setup uses a signal modulation at a frequency f_{mod} = 1 MHz which is far larger than the resonator mechanical bandwidth. The amplitude of the test signal has a value of $\approx 250\ mV_{eff}$. Measurements are controlled through *Labview* programs.

The devices were *wafer-wafer* packaged at a pressure $p \approx 1$ mbar, and the pads have been wire bonded to an holding board which can be tilted and which is electrically connected to the main, fixed, readout board. The actual resonators have a length L = 400 μm, a thickness t = 15 μm and a nominal width h = 1.2 μm.

A. Stationary response

Fig. 4 reports experimental results representing the sensed capacitance variation (measured between the sensing plate and the resonator) with respect to a stationary voltage applied between the driving plate and the resonator. As the driving voltage increases, the beam is deflected towards the actuation plate and as a consequence the sensing capacitance decreases

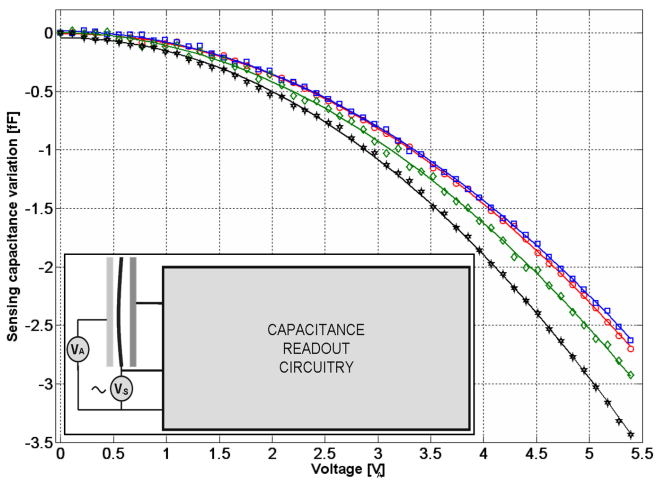


Fig. 4. Sensing capacitance vs driving actuator voltage curves for 4 different resonators belonging to 2 different accelerometers (experimental points and fitting quadratic polynominial). A certain data spreading is the result of the high shrinkage used in the resonator width design. Measurements are performed applying a voltage V_A to the driving electrode and measuring the sensing capacitance through a small-amplitude, high frequency signal V_S, processed through a custom lock-in type readout electronics.

of $\approx$ 3 fF at 5.4 V. In the simplified assumption of a parallel plate driving capacitor of area A_S and gap x_0, the theoretical relationship between the applied driving voltage V_A and the applied electrostatic force F_{el} is a function of the resonator displacement x:

$$F_{el}(V_A, x) = \frac{d}{dx}[\frac{V_A^2 \cdot \epsilon_0 \cdot A_S}{2(x_0 - x)}] = \frac{V_A^2}{2}\frac{C_0}{x_0} \cdot \frac{1}{(1 - x/x_0)^2} \quad (4)$$

being C_0 the rest capacitance value (for x = 0) and ϵ_0 the air permittivity (taken as that of vacuum). For small displacements ($x \rightarrow 0$) the force can be assumed to be independent from the position:

$$F_{el}(V_A) = \frac{V_A^2}{2}\frac{C_0}{x_0} \quad (5)$$

The beam displacement x is proportional through the elastic stiffness k_{beam} of the beam to the applied force F_{el}:

$$x = F_{el}/k_{beam} \propto V_A^2 \quad (6)$$

and the measured sensing capacitance variation ΔC_S caused by a displacement x is, in the first order approximation of small displacements:

$$\Delta C_S(x) = \frac{\epsilon_0 A_S}{x_0} - \frac{\epsilon_0 A_S}{x_0 + x} \approx -C(0)\frac{x}{x_0} \propto -V_A^2 \quad (7)$$

being A_S the area of the sensing capacitor (which is the same as the area of the driving capacitor in our device).

It is worthnoting that in the present case the resonating beam deforms and therefore the gap variation is non-uniform. However the above formula can still be used, with a proper correction coefficient, by interpreting x as the displacement of the beam mid point.

It turns thus out that the sensing capacitance variation, which is proportional to the resonator displacement x, has a square

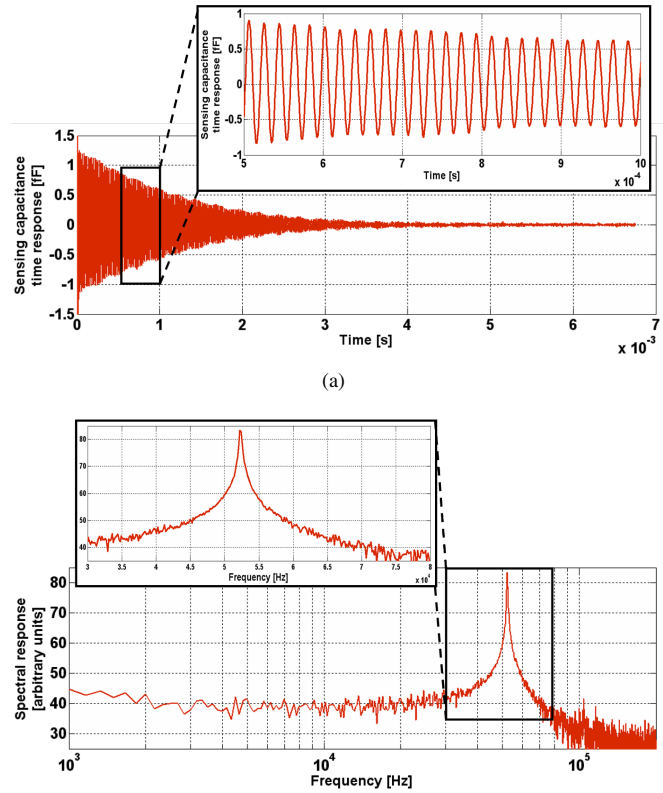


Fig. 5. (a) Time response to a downward force step (to the rest position) of a single resonator of the accelerometer. (b) Corresponding spectral response derived through the DFT.

dependence from the driving voltage V_A for small displacements of the resonator. Indeed parabolic curves in Fig. 4 well fit the experimental points, as the displacements are small with respect to the gap between the resonator and the driving electrode. The results show a non negligible data spreading, caused by the very reduced width of the resonator, which is thus affected by an unavoidabe process dispersion.

B. Dynamic response

In this subsection the experimental results representing the dynamic response of the resonators are reported. The tests have been carried out in the following way: through a quasi-stationary voltage V_A applied between the driving plate and the resonator, the beam is slowly deflected until an established perturbed position. Then the voltage V_A is instantaneously set to zero. As a consequence of this downward electrostatic force step, the resonator begins to oscillate around the rest position and through the sensing plate the value of the sensing capacitance $C_S(t)$ is real-time monitored, as reported in Fig. 5 (a).

From this measurement the spectral response of the device can be calculated by means of the Discrete Fourier Transform. Fig. 5 (b) reports a typical result, showing a resonance frequency f_0 = 52.21 kHz, with a quality factor $Q \approx 208$.

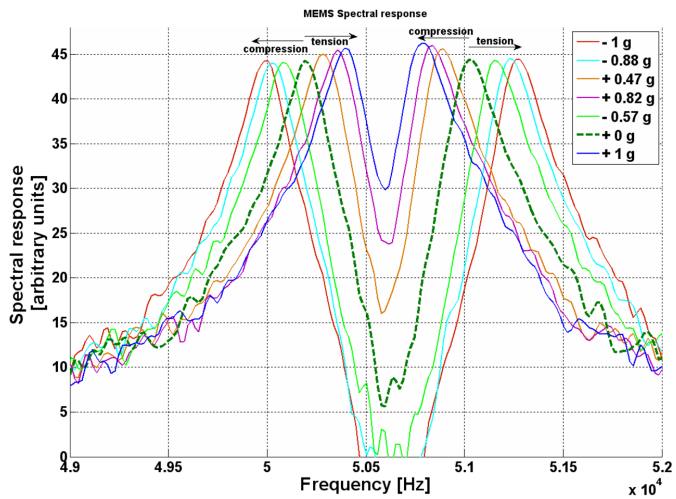


Fig. 6. Peak frequency shift of the resonators of the resonant accelerometer. The dashed curve represents the response at 0 g (where a small difference in the resonance frequency is the result of a process data spreading). The solid curves represent the spectral response at different external acceleration in the range $\pm$ 1 g.

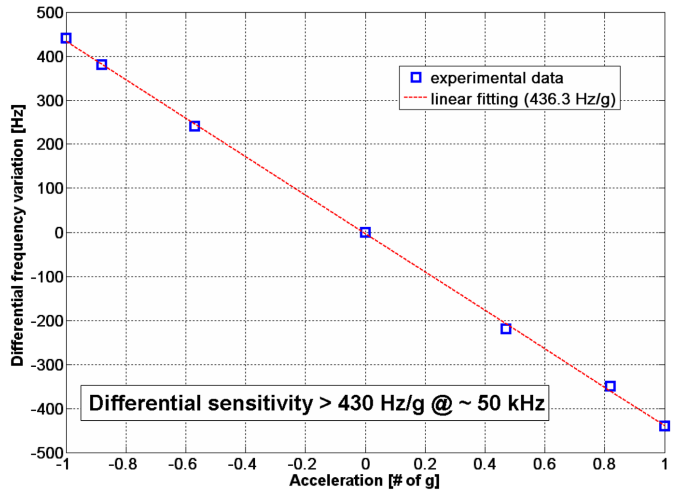


Fig. 7. Variation of the peak frequency difference between the resonators $\Delta f - \Delta f_0$ as a function of the external acceleration in the range $\pm$ 1 g. Δf_0 corresponds to the peak frequency difference at 0 g. The device shows a good linearity and a sensitivity higher than 430 Hz/g.

IV. Experimental Response to External Accelerations

The response of the device to external accelerations has been measured in the range $\pm$ 1 g. The variation of the external applied acceleration is simply obtained through different inclinations of the device. The resonance frequency is acquired, as described in section III B, at different tilt angles between -90° and $+90^o$. To simultaneously acquire the data from both the resonators, the driving and sensing pads of the two resonators have been short circuited. Fig. 6 reports the expermimental results obtained acquiring simultaneously the data from the two sensing capacitances. The spectral response presents a typical double peak, due to the fact that the resonance frequencies of the two resonators are not equal, as a consequence of the process spreading already observed in the CV measurements.The results show that when a resonator is compressed (and the other one is at the same time stretched) its resonance frequency decreases (and that of the other resonator increases).

The device linearity (see Eq. 3) is reported in Fig. 8. Being Δf_0 the frequency difference between the resonant peaks of the two resonators at 0 g, the experimental points represent the variation of the peak frequency difference $\Delta f - \Delta f_0$ as a function of the external acceleration in the range $\pm$ 1 g. A good linearity and an overall differential sensitivity larger than 430 Hz/g (at a resonance frequency of both resonators around 50 kHz) is obtained.

Fig. 8 reports a SEM picture of an unpackaged device. A detail of the point where the resonator is attached to the spring is shown in the close up.

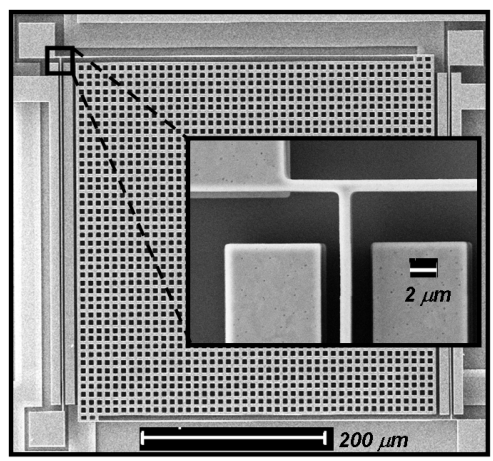


Fig. 8. Scanning Electron Microscope image of an unpackaged device. The detail shows the point where the resonator is attached to the spring. This point is chosen to have the maximum device sensitivity.

V. Conclusion

First experimental results on an innovative uniaxial resonant accelerometer have been presented. The device has an overall area occupation lower than (500 μm^2). Thanks to an optimization of the geometrical parameters, a very high differential sensitivity, close to 500 Hz/g (around a resonance frequency of 50 kHz) has been obtained, with a good linearity in the range $\pm$ 1 g. The work in progress includes the re-design of the resonating parts of the accelerometers, to reduce a certain performance spreading caused by an underestimation of the process overetch, and the coupling of the device to a suitable readout electronics. A two dimensional accelerometer based on the same principle is also under development.

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